Quadratic equations using technology

1 Solve for the pronumeral for each of the following quadratic equations:

a $4m\left(m+5\right)=0$

b $\frac{m}{2}\left(m+5\right)=0$

c $-25v^2+64=0$

d $\frac{x^2}{16}-2=2$

e $81k^2+8=24$

f $10\left(p^2-7\right)=930$

g $4y^2=100$

h $25y^2=36$

i $-3k^2=-12$

2 Solve for x for each of the following quadratic equations:

a $\left(8x+9\right)^2=256$

b $\left(x-4\right)\left(x-2\right)=0$

c $x\left(x+7\right)=0$

d $\left(8x-5\right)\left(3x-7\right)=0$

e $\left(3x+8\right)\left(5x-7\right)=0$

f $\left(3x-17\right)\left(2x+d\right)=0$

g $x^2=294$

h $81x^2-16=0$

i $\left(10x-9\right)^2=0$

j $\left(-3+7x\right)^2=0$

k $x\left(2x-9\right)=0$

l $\left(x-6\right)\left(x+7\right)=0$

m $x^2=2$

n $\left(x-3\right)^2=64$

o $\left(x+3\right)^2=49$

p $\left(2-x\right)^2=81$

q $\left(x-6\right)^2=2$

r $x^2=25$

s $x^2=121$

t $x^2-121=0$

u $x^2-10=15$

3 Solve the following equations by using technology. Round your answers to one decimal place where necessary.

a $3x^2-x-10=0$

b $x^2+2x-\frac{21}{4}=0$

c $4.6x^2+7.3x-3.7=0$

4 Consider the equation $3x^2=6$.

a Solve the equation by using technology, giving your answers in exact form.

b Give the solutions as decimals rounded to the nearest tenth.

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5 Consider the equation $0 = -x^2 + 2x - 1$. Use technology to solve the equation for x.

a Write down how many solutions the equation has.

b Hence, state how many x-intercepts there are on the graph of the function

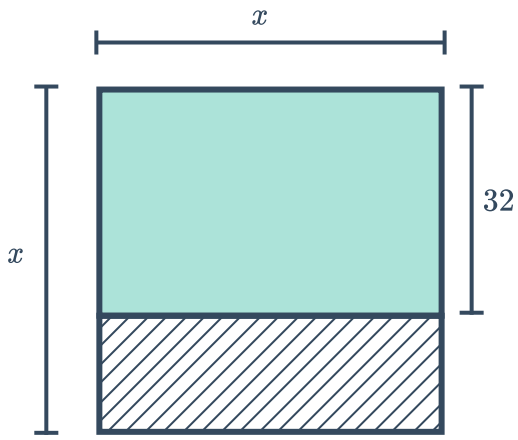
$$y = -x^2 + 2x - 1$$

6 Solve $\left(5x^2 + 13x + 6\right)\left(2x^2 + 13x + 20\right) = 0$.

Applications

7 The Widget and Trinket Emporium has released the forecast of its revenue over then next year. The revenue R (in dollars) at any point in time t (in months) is described by the equation $R = -(t - 12)^2 + 4$. Solve the equation $-(t - 12)^2 + 4 = 0$ to find the times at which the revenue will be zero.

8 Neville needs a sheet of paper x cm by 13 cm for an origami giraffe. The local origami supply store only sells square sheets of paper.
The lower portion of the image shows the excess area A of paper that will be left after Neville cuts out the x cm by 32 cm piece.
The excess area, in cm^2, is given by the equation $A = x(x - 32)$.

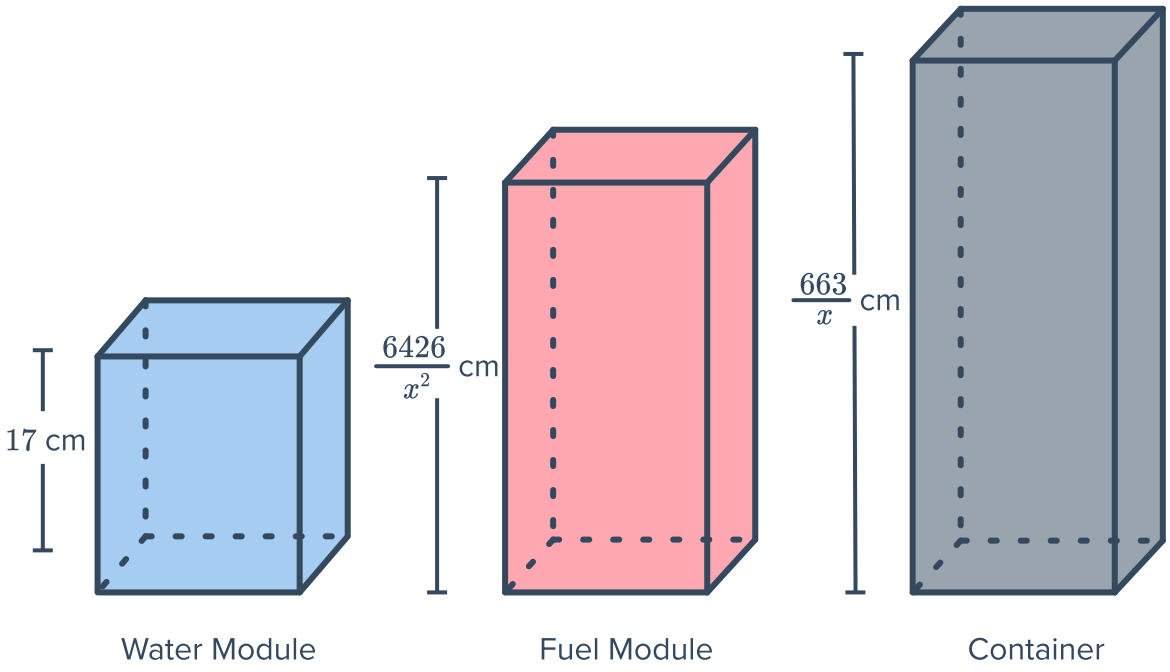


a At what lengths, x, will the excess area be zero?

b For what value of x will Neville be able to make an origami giraffe with the least amount of excess paper?

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9 An interplanetary freight transport company has won a contract to supply the space station orbiting Mars. They will be shipping stackable containers, each carrying a fuel module and a water module, that must meet certain dimension restrictions.
The design engineers have produced a sketch for the modules and container, shown below.
The sum of the heights of both modules equal to the height of the container.



a Write an equation that equates the height of the container and the sum of the heights of the modules.

b Find the possible values of x.

c Find the tallest possible height of the container. Give your answer to two decimal places.

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